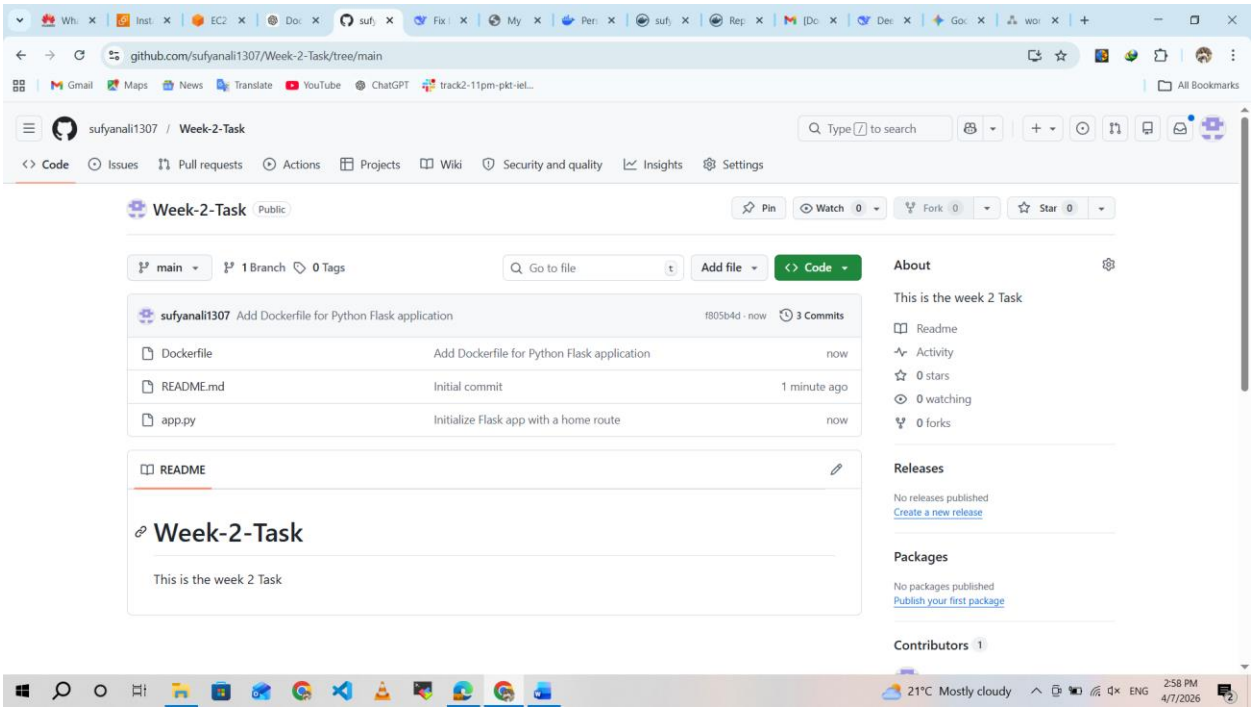
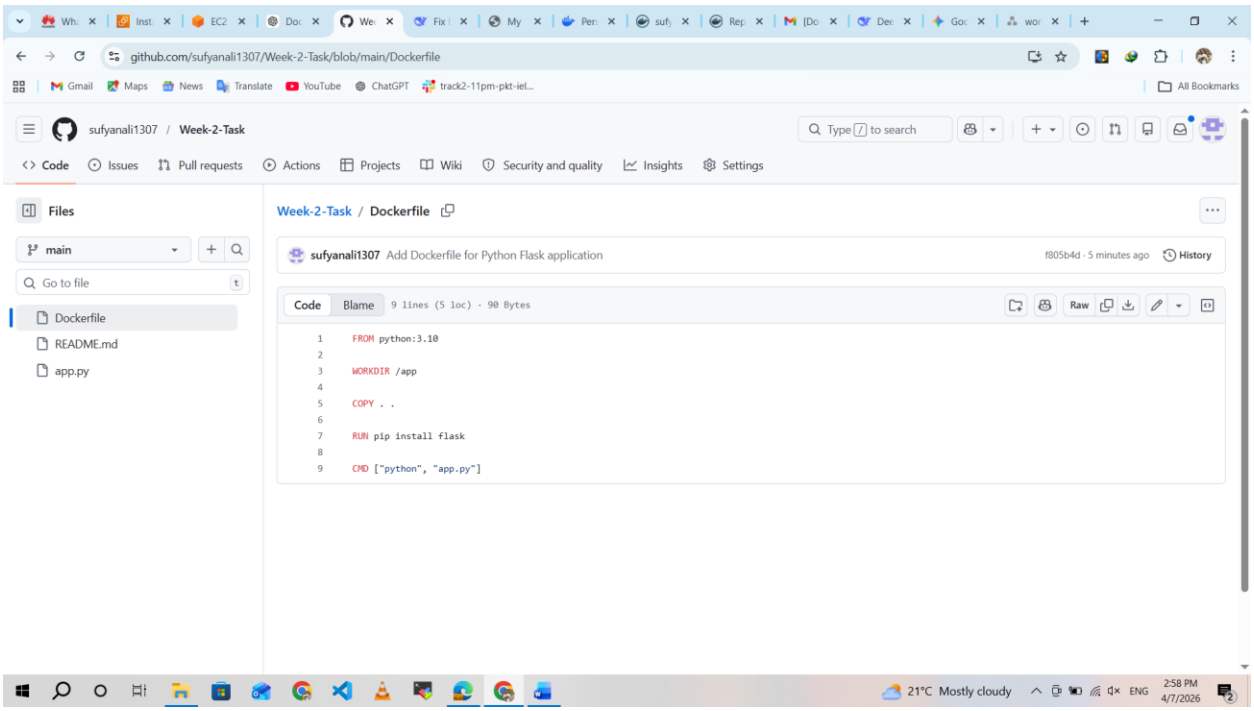
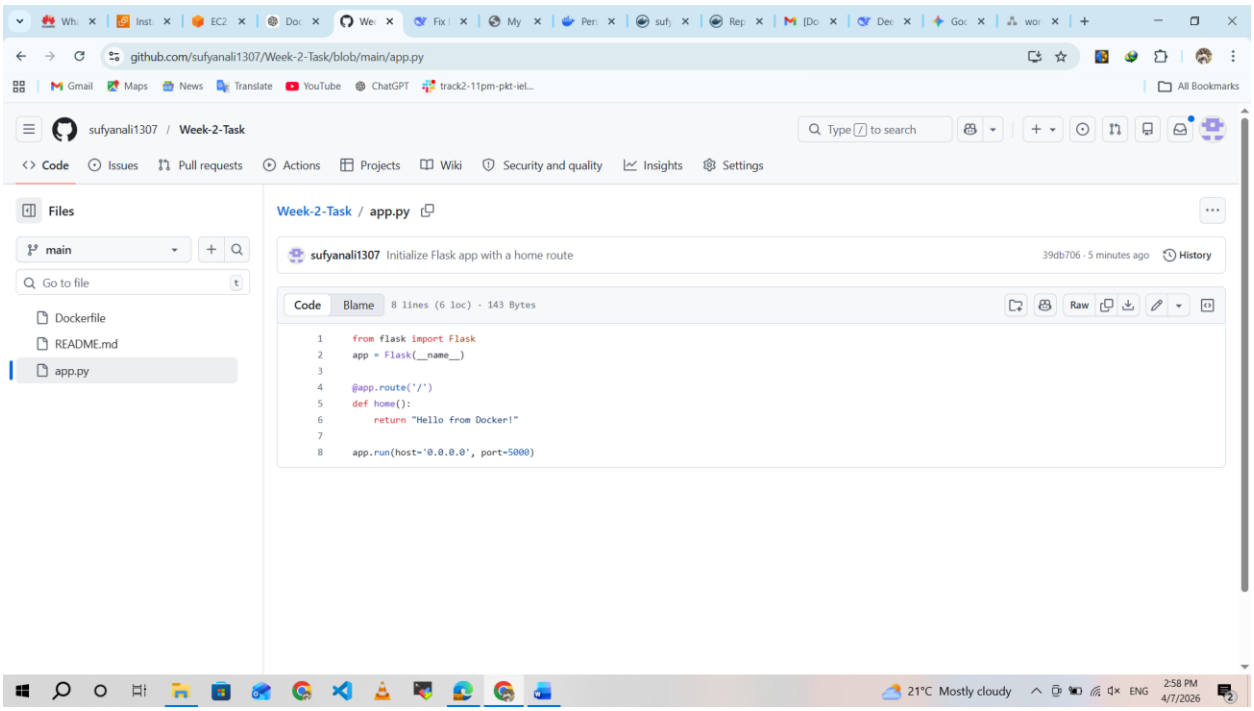
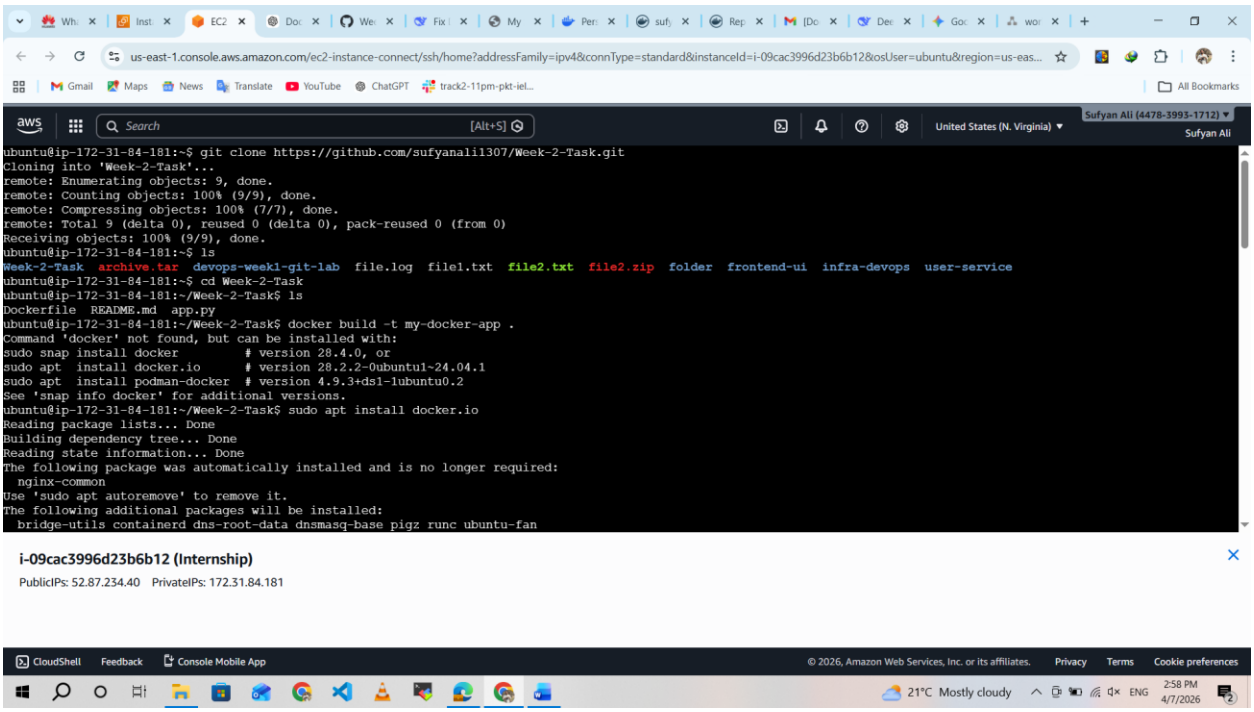
Week 2

HANDS-ON LAB #3 — Containerize a Real App






github.com/sufyanali1307/Week-2-Task/blob/main/app.py
sufyanali1307 / Week-2-Task
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
Dockerfile
README.md
app.py
Week-2-Task / app.py
sufyanali1307 Initialize Flask app with a home route
39db706 · 5 minutes ago
History
Code
Blame
8 lines (6 loc) · 143 Bytes
Raw
from flask import Flask
app = Flask(__name__)
@app.route('/')
def home():
return "Hello from Docker!"
app.run(host='0.0.0.0', port=5000)


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-09cac3996d23b6b12&osUser=ubuntu&region=us-eas...
United States (N. Virginia)
Sufyan Ali (4478-3993-1712)
Sufyan Ali
ubuntu@ip-172-31-84-181:~$ git clone https://github.com/sufyanali1307/Week-2-Task.git
Cloning into 'Week-2-Task'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
ubuntu@ip-172-31-84-181:~$ ls
Week-2-Task archive.tar devops-week1-git-lab file.log file1.txt file2.txt file2.zip folder frontend-ui infra-devops user-service
ubuntu@ip-172-31-84-181:~$ cd Week-2-Task
ubuntu@ip-172-31-84-181:~/Week-2-Task$ ls
Dockerfile README.md app.py
ubuntu@ip-172-31-84-181:~/Week-2-Task$ docker build -t my-docker-app .
Command 'docker' not found, but can be installed with:
sudo snap install docker # version 28.4.0, or
sudo apt install docker.io # version 28.2.2-0ubuntu1~24.04.1
sudo apt install podman-docker # version 4.9.3+ds1-1ubuntu0.2
See 'snap info docker' for additional versions.
ubuntu@ip-172-31-84-181:~/Week-2-Task$ sudo apt install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
nginx-common
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
i-09cac3996d23b6b12 (Internship)
PublicIPs: 52.87.234.40 PrivateIPs: 172.31.84.181
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-09cac3996d23b6b12&osUser=ubuntu&region=us-eas...

Sufyan Ali (4478-3993-1712) · United States (N. Virginia)

```
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-84-181:~/Week-2-Task$ docker build -t my-docker-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

time="2026-04-07T09:56:23Z" level=error msg="Can't add file /home/ubuntu/Week-2-Task/.git to tar: io: read/write on closed pipe"
time="2026-04-07T09:56:23Z" level=error msg="Can't close tar writer: io: read/write on closed pipe"
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/build?dockerfile=Dock
erfile&t=my-docker-app&version=1": dial unix /var/run/docker.sock: connect: permission denied
ubuntu@ip-172-31-84-181:~/Week-2-Task$ sudo usermod -aG docker $USER
ubuntu@ip-172-31-84-181:~/Week-2-Task$ newgrp docker
ubuntu@ip-172-31-84-181:~/Week-2-Task$
```

i-09cac3996d23b6b12 (Internship)

PublicIPs: 52.87.234.40 PrivateIPs: 172.31.84.181

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-09cac3996d23b6b12&osUser=ubuntu&region=us-eas...

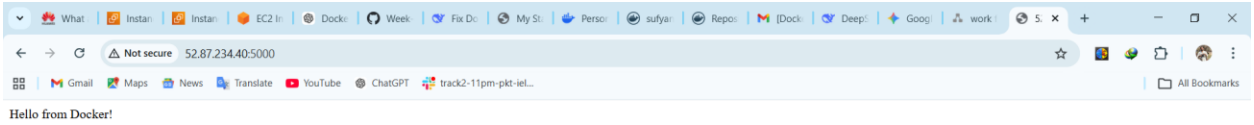
Sufyan Ali (4478-3993-1712) · United States (N. Virginia)

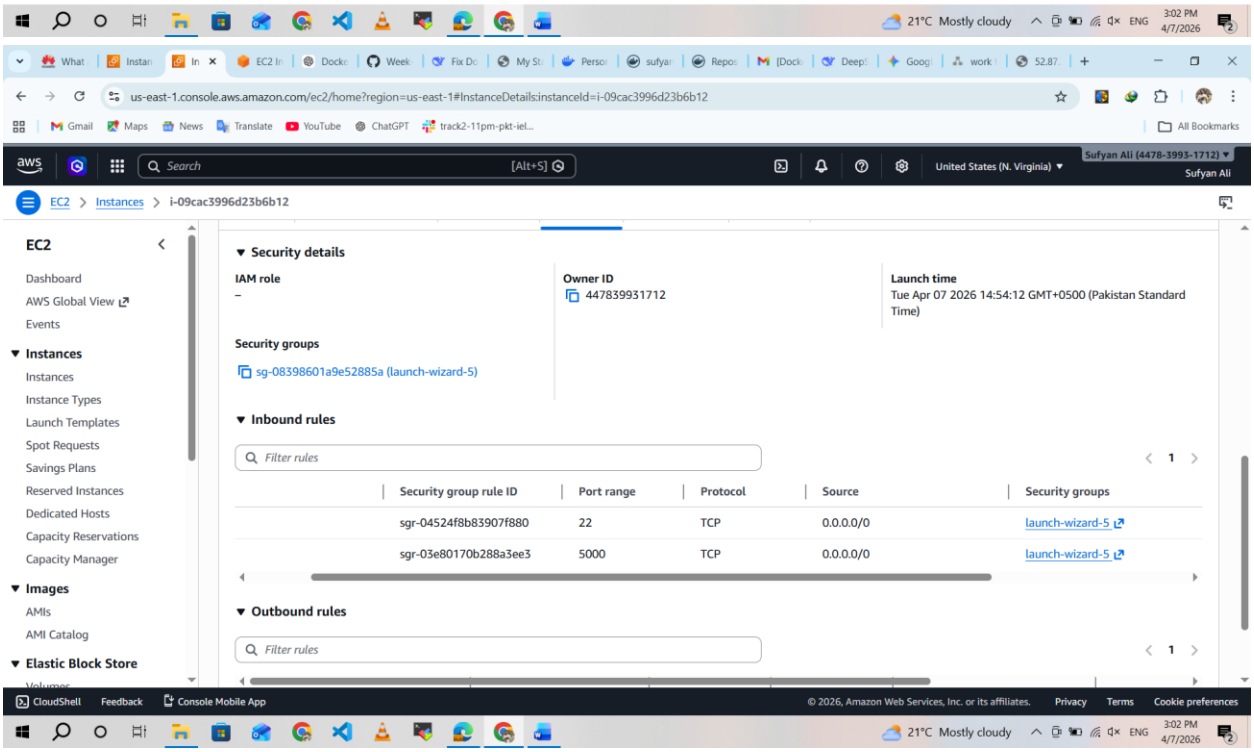
```
Collecting jinja2>=3.1.2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 kB 18.4 MB/s eta 0:00:00
Collecting werkzeug>=3.1.0
  Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 226.5/226.5 kB 28.1 MB/s eta 0:00:00
Collecting itsdangerous>=2.2.0
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting click>=8.1.3
  Downloading click-8.3.2-py3-none-any.whl (108 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 108.4/108.4 kB 17.7 MB/s eta 0:00:00
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeug, jinja2, flask
Successfully installed blinker-1.9.0 click-8.3.2 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.8
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual en
vironment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
 ---> Removed intermediate container 1feac4df6382
 ---> 89e350247007
Step 5/5 : CMD ["python", "app.py"]
 ---> Running in 2d593aac39f1
 ---> Removed intermediate container 2d593aac39f1
 ---> c9bb83705a30
Successfully built c9bb83705a30
Successfully tagged my-docker-app:latest
ubuntu@ip-172-31-84-181:~/Week-2-Task$
```

i-09cac3996d23b6b12 (Internship)

PublicIPs: 52.87.234.40 PrivateIPs: 172.31.84.181


Not secure
52.87.234.40:5000
Hello from Docker!


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-09cac3996d23b6b12
United States (N. Virginia)
Sufyan Ali (4478-3993-1712)
EC2 > Instances > i-09cac3996d23b6b12
Security details
IAM role
–
Owner ID
447839931712
Launch time
Tue Apr 07 2026 14:54:12 GMT+0500 (Pakistan Standard Time)
Security groups
sg-08398601a9e52885a (launch-wizard-5)
Inbound rules
Security group rule ID
Port range
Protocol
Source
Security groups
sgr-04524f8b83907f880
22
TCP
0.0.0.0/0
launch-wizard-5
sgr-03e80170b288a3ee3
5000
TCP
0.0.0.0/0
launch-wizard-5
Outbound rules
Filter rules
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 108.4/108.4 kB 17.7 MB/s eta 0:00:00
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeug, jinja2, flask
Successfully installed blinker-1.9.0 click-8.3.2 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.8
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 ---> Removed intermediate container 1feac4df6382
 ---> 89e350247007
Step 5/5 : CMD ["python", "app.py"]
 ---> Running in 2d593aac39f1
 ---> Removed intermediate container 2d593aac39f1
 ---> c9bb83705a30
Successfully built c9bb83705a30
Successfully tagged my-docker-app:latest
ubuntu@ip-172-31-84-181:~/Week-2-Task$ docker run -p 5000:5000 my-docker-app
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
119.155.140.234 - - [07/Apr/2026 10:02:15] "GET / HTTP/1.1" 200 -
119.155.140.234 - - [07/Apr/2026 10:02:16] "GET /favicon.ico HTTP/1.1" 404 -
```

i-09cac3996d23b6b12 (Internship)

PublicIPs: 52.87.234.40 PrivateIPs: 172.31.84.181

```
[notice] A new release of pip is available: 23.0.1 -> 26.0.1
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 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
119.155.140.234 - - [07/Apr/2026 10:02:15] "GET / HTTP/1.1" 200 -
119.155.140.234 - - [07/Apr/2026 10:02:16] "GET /favicon.ico HTTP/1.1" 404 -
^Cubuntu@ip-172-31-84-181:~/Week-2-Taskdocker tag my-docker-app sufyanali1307/my-docker-app:1.0
ubuntu@ip-172-31-84-181:~/Week-2-Task$ docker images
REPOSITORY                  TAG       IMAGE ID       CREATED          SIZE
my-docker-app               latest    c9bb83705a30   5 minutes ago    1.11GB
sufyanali1307/my-docker-app 1.0       c9bb83705a30   5 minutes ago    1.11GB
python                      3.10      738f70a7db01   5 hours ago      1.09GB
ubuntu@ip-172-31-84-181:~/Week-2-Task$
```
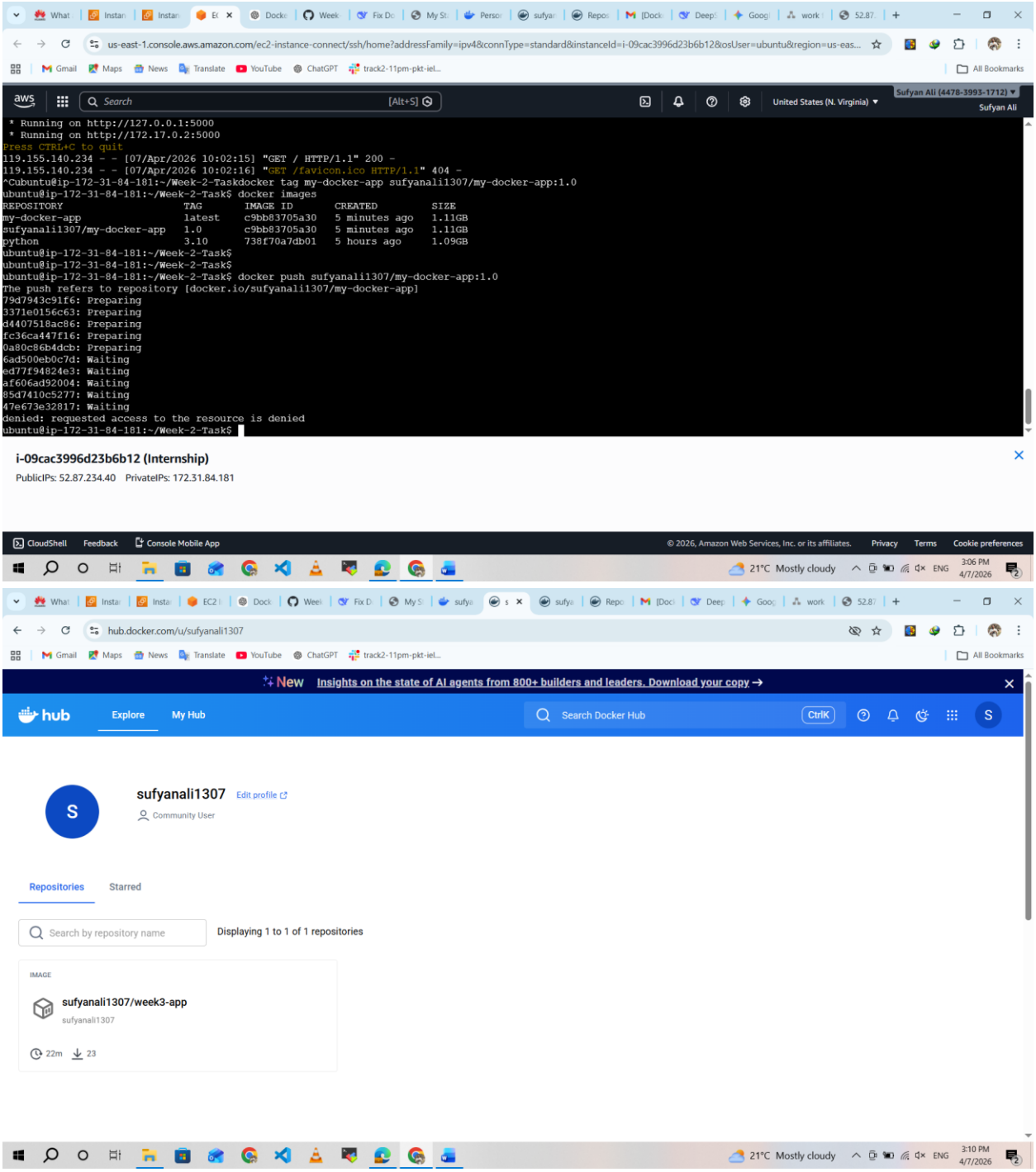

i-09cac3996d23b6b12 (Internship)

PublicIPs: 52.87.234.40 PrivateIPs: 172.31.84.181



Docker hub Link

https://hub.docker.com/u/sufyanali1307?_gl=1*akcwyf*_gcl_au*MTg3MzgwNDc2Ni4xNzc1NTM4Nzc0*_ga*ODY5MjExMDk4LjE3NjgyOTIzMDU.*_ga_XJWPQMJYHQ*czE3NzU1NTY0NTEkbzMkZzEkdDE3NzU1NTY2NDMkajYkbDAkaDA.

Git hub Repo link

https://github.com/sufyanali1307/Week-2-Task

What the image contains?

A Docker image contains the application code, all required dependencies and libraries, and a base operating system needed to run the app. It also includes configuration files and instructions (from the Dockerfile) that define how the application should start and run inside the container.

How Docker networking works?

Docker networking works by connecting containers to networks so they can communicate with each other and the outside world. It uses port mapping (like -p 5000:5000) to link a container's port to the host system, allowing access through a browser. Docker also provides internal networks where containers can talk to each other using names instead of IP addresses.

Why containers are important?

Containers are important because they provide a consistent and isolated environment for applications, ensuring they run the same on any system. They are lightweight and fast, use fewer resources than virtual machines, and make deployment, scaling, and maintenance easier across development, testing, and production environments.